

Database Management Systems

Lecture 2

Data Abstraction

Suppression of details of data organization and storage, and highlighting of the essential features for an improved understanding of data.

One of the main characteristics of database approach is to support data abstraction so that different users can perceive data at their preferred level of detail.

****Hiding low-level storage/implementation details and showing users only what they need.**

Levels of Abstraction

3 Levels

External (View) Level (Highest)

Logical/Conceptual Level

Physical Level (Lowest)

Data Model

To represent data at each of the 3 levels, we need some language or notations to express the structures.

At the Logical/Conceptual level, the data model we use is the ER(Entity Relationship) model.

Entity Relationship Model (ER Model)

Uses concepts like:

- Entity
- Attribute
- Relationship

(We will study ER models in the next Chapter)

Schema

The actual design/description of a database built using the data model.

Client/Server Architectures

A computing model where work is divided between 2 types of machines:

Client: The front-end that requests a service

Server: The back-end provides the service

The Server processes the request and sends it back to the client.

Client/Server Architectures in DBMS

Separation of concerns. Dividing a system into parts, each responsible for a specific function, so they can work independently without one interfering the other.

Advantages:

Maintainability: UI developers can change the UI without touching the database

Scalability: Database servers focus on queries, not rendering screens.

Flexibility: Multiple UIs (desktop app, mobile app, web app) can reuse the same DB.

Security: Servers control access, clients don't touch raw storage.

Two-Tier Client/Server Architecture

Client (Tier 1) ↔ Database Server (Tier 2)

Client app sends SQL directly to DBMS

DBMS executes and sends results back

Three-Tier Client/Server Architecture

Client (Tier 1) ↔ Application Server (Tier 2) ↔ Database Server (Tier 3)

Adds an extra layer for **business logic**, making it more **scalable**, secure and perfect for web applications.